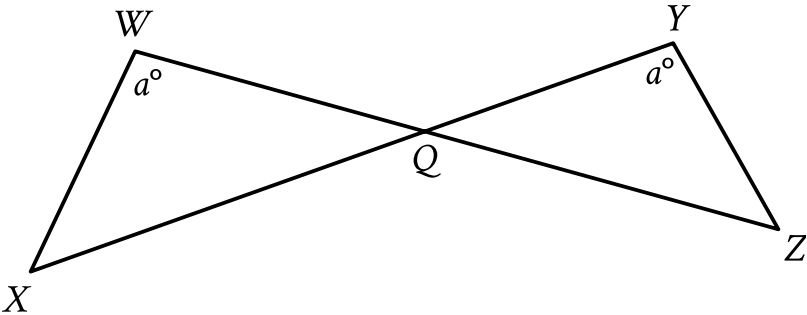


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

Answer

- A. 18
- B. 36
- C. 120
- D. 200

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of a° , so they are congruent to each other. Therefore, $\triangle WQX$ and $\triangle YQZ$ are similar triangles, where $\overline{WQ}$ corresponds to $\overline{YQ}$ and $\overline{WX}$ corresponds to $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of $\overline{YZ}$ is 18.

Choice B is incorrect. This is the length of $\overline{QZ}$, not $\overline{YZ}$.

Choice C is incorrect. This is the length of $\overline{XQ}$, not $\overline{YZ}$.

Choice D is incorrect and may result from conceptual or calculation errors.